True: normal
1st: normal (84.7%)
2nd: tuberculosis (10.2%)

True: normal
1st: tuberculosis (54.9%)
2nd: normal (44.3%)

True: normal
1st: tuberculosis (51.5%)
2nd: normal (47.6%)

True: normal
1st: tuberculosis (57.8%)
2nd: normal (41.1%)

True: normal
1st: normal (63.5%)
2nd: tuberculosis (33.6%)

True: normal
1st: pneumonia (59.3%)
2nd: normal (40.6%)

True: normal
1st: normal (54.3%)
2nd: pneumonia (45.5%)

True: normal
1st: tuberculosis (50.9%)
2nd: normal (47.2%)

True: normal
1st: normal (53.9%)
2nd: tuberculosis (44.7%)

True: normal
1st: normal (79.2%)
2nd: pneumonia (19.5%)

True: normal
1st: pneumonia (56.8%)
2nd: normal (42.8%)

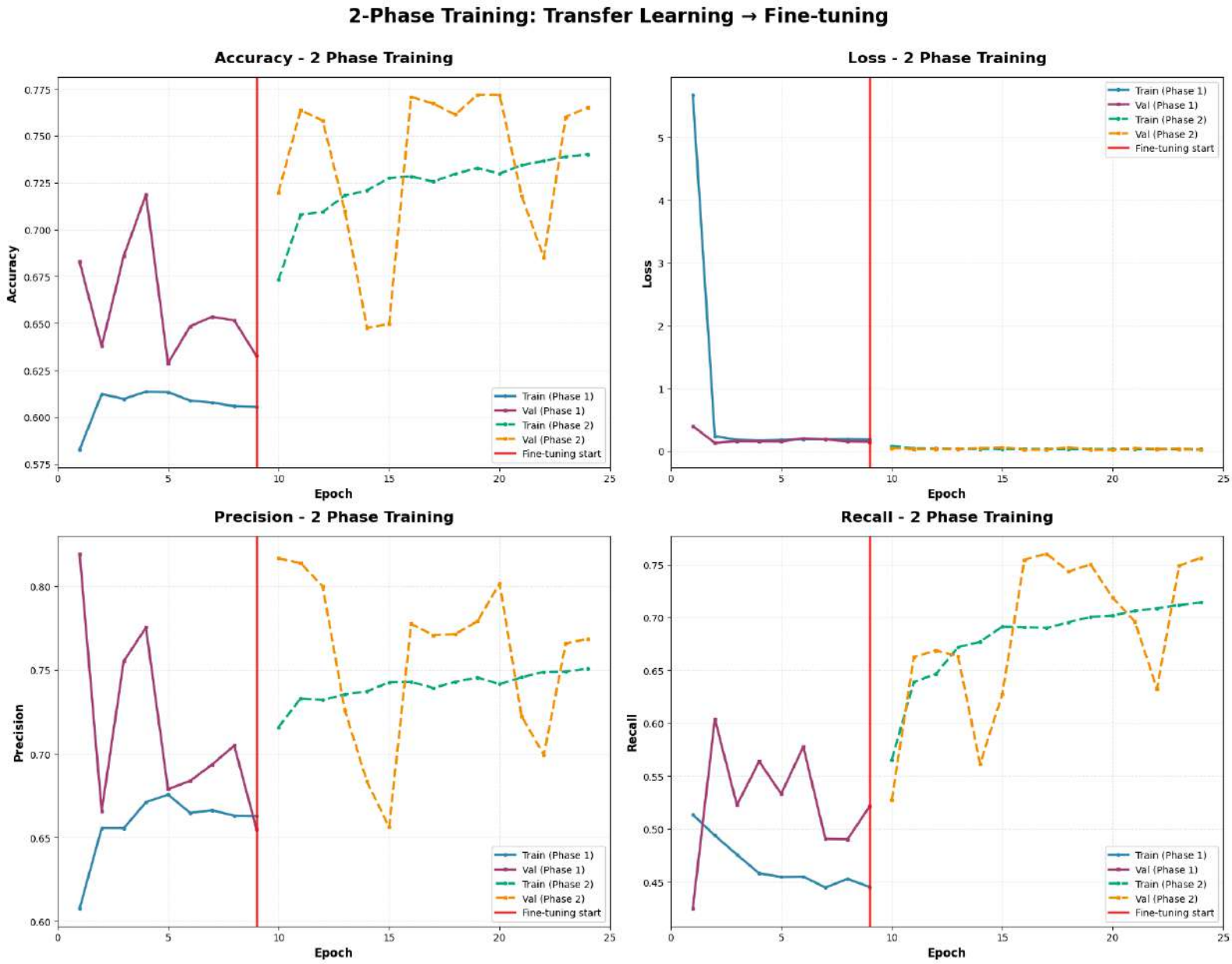
True: normal
1st: normal (52.2%)
2nd: tuberculosis (46.5%)

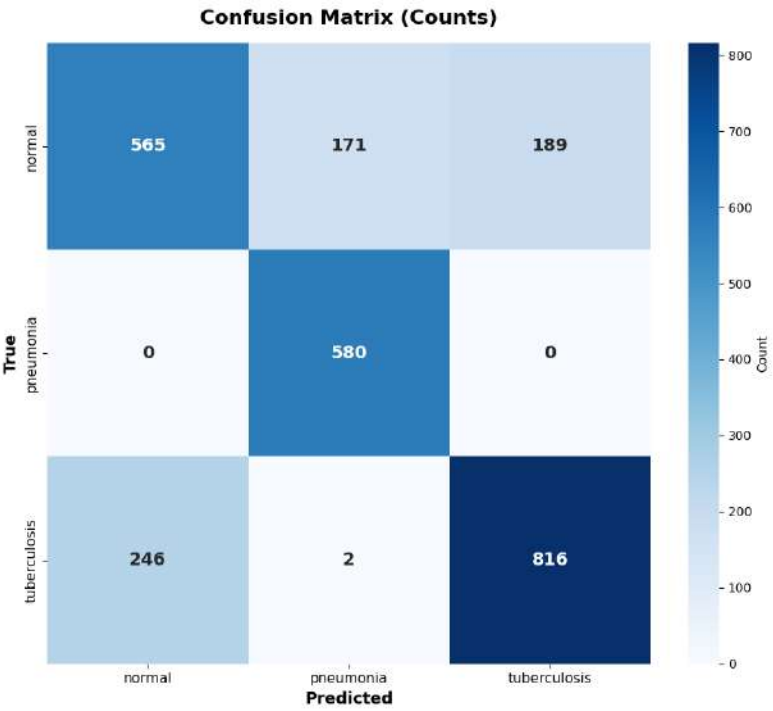
True: normal
1st: normal (50.3%)
2nd: tuberculosis (48.6%)

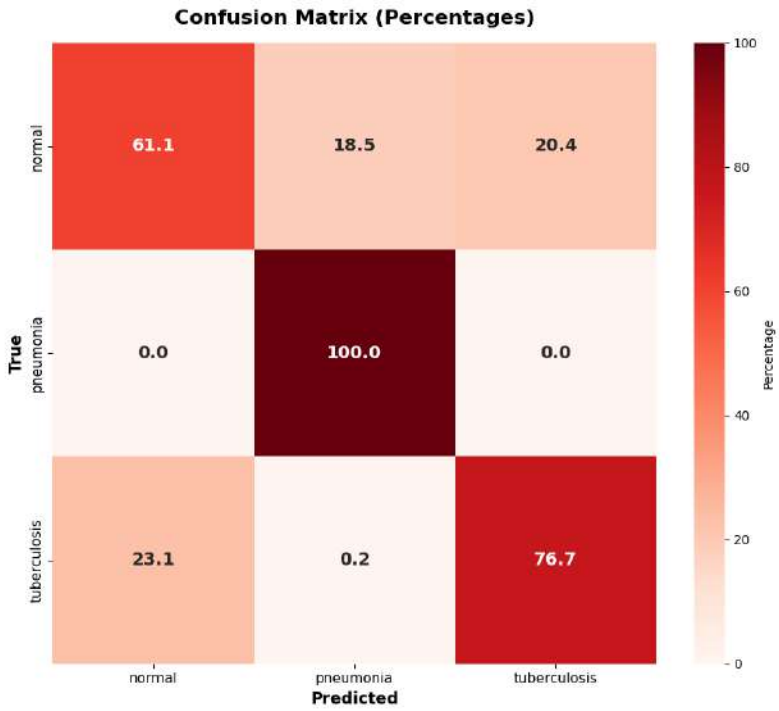
True: normal
1st: pneumonia (58.5%)
2nd: normal (41.2%)

True: normal
1st: normal (58.5%)
2nd: tuberculosis (39.5%)

True: normal
1st: normal (65.2%)
2nd: tuberculosis (32.4%)


2-Phase Training: Transfer Learning → Fine-tuning
Accuracy - 2 Phase Training
Accuracy
Epoch
Loss - 2 Phase Training
Loss
Epoch
Precision - 2 Phase Training
Precision
Epoch
Recall - 2 Phase Training
Recall
Epoch
Train (Phase 1)
Val (Phase 1)
Train (Phase 2)
Val (Phase 2)
Fine-tuning start


Confusion Matrix (Counts)
True
normal
pneumonia
tuberculosis
565
171
189
0
580
0
246
2
816
Predicted
normal
pneumonia
tuberculosis
Count
800
700
600
500
400
300
200
100
0


Confusion Matrix (Percentages)
True
normal
pneumonia
tuberculosis
61.1
18.5
20.4
0.0
100.0
0.0
23.1
0.2
76.7
Predicted
normal
pneumonia
tuberculosis
Percentage
100
80
60
40
20
0